Chapter 2

Complex Differentiation

2.1 Continuity of complex functions

Definition 2.1. A function $f : \mathcal{D} \to \mathbb{R}^q$ with $\mathcal{D} \subseteq \mathbb{R}^p$ is said to be *continuous at a point* $a \in \mathcal{D}$ iff

$$\begin{aligned} \forall \epsilon > 0, \quad & \exists \delta > 0 \text{ s.t. } \forall z \in \mathcal{D} \setminus \{a\}, \\ & |z - a| < \delta \implies |f(z) - f(a)| < \epsilon, \end{aligned} \tag{2.1}$$

where $|\cdot|$ denotes the Euclidean norm. f is *continuous* on $\mathcal{D}$ iff it is continuous at each point $a \in \mathcal{D}$.

Lemma 2.2. A function $f : \mathcal{D} \to \mathbb{R}^q$ with $\mathcal{D} \subseteq \mathbb{R}^p$ is continuous at a if and only if $\lim_{n\to+\infty} f(a_n) = f(a)$ for *each* sequence (a_n) that converges to a.

Proof. This follows from Definitions 2.1 and B.3. □

Lemma 2.3. The sum, difference, and product of two continuous functions are continuous.

Proof. This follows from Definition 2.1. □

Example 2.4. A complex polynomial of degree $n \in \mathbb{N}$

$$p_n(z) = \sum_{j=0}^{n} a_j z^j, \qquad a_j \in \mathbb{C}. \tag{2.2}$$

is continuous.

Lemma 2.5. A function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ is continuous if and only if its real and imaginary parts

$$(\operatorname{Re} f)(z) := \operatorname{Re} f(z), \qquad (\operatorname{Im} f)(z) := \operatorname{Im} f(z) \tag{2.3}$$

are continuous.

Proof. This follows from Definition 2.1 and Corollaries 1.64 and 1.66. □

Corollary 2.6. The modulus of a continuous complex function is continuous.

Proof. This follows from Definition 1.63, Lemma 2.5 and Definition 2.1. □

Corollary 2.7. The function $f(z) = z^{-1}$ is continuous.

Proof. This follows from Corollary 1.65, Definition 1.61, Lemma 2.5, and Corollary 2.6. □

Lemma 2.8. The composition of two continuous functions are continuous.

Proof. This follows from Definition 2.1. □

Corollary 2.9. If $f : \mathcal{D} \to \mathbb{C}^\bullet$ is continuous, so is the function $\frac{1}{f} : \mathcal{D} \to \mathbb{C}^\bullet$.

Proof. The conclusion follows from setting $g(z) = z^{-1}$ and applying Corollary 2.7 and Lemma 2.8 to $g \circ f$. □

Definition 2.10. The *inverse function* of an injection $f : \mathcal{D} \to \mathbb{C}$ is a function $f^{-1} : f(\mathcal{D}) \to \mathcal{D}$ satisfying

$$\forall z \in D, \qquad f^{-1}(f(z)) = z, \tag{2.4a}$$

$$\forall w \in f(\mathcal{D}), \qquad f(f^{-1}(w)) = w. \tag{2.4b}$$

Lemma 2.11. The function $z \mapsto \operatorname{Arg} z$ is not continuous along the negative real axis.

Proof. By Definition 2.1, we need to show that the negation of (2.1) holds with $a = -1$, i.e.,

$$\begin{aligned} \exists \epsilon > 0, \text{ s.t. } \forall \delta > 0, \ \exists z \in \mathbb{C}^\bullet, \text{ s.t.} \\ |z + 1| < \delta \implies |\operatorname{Arg}(z) - \operatorname{Arg}(a)| \geq \epsilon, \end{aligned}$$

which holds by setting $\epsilon = \frac{\pi}{2}$ and $z = -1 - \mathbf{i}\frac{\delta}{2}$. □

Exercise 2.12. The function $z \mapsto \operatorname{Arg} z$ is continuous on the complex plane slit along the negative real axis

$$\mathbb{C}_- := \mathbb{C} \setminus \{t \in \mathbb{R} : t \leq 0\}. \tag{2.5}$$

Corollary 2.13. The principal value of the logarithm is discontinuous along the negative real axis. However, it is continuous on $\mathbb{C}_-$.

2.2 Complex derivatives

Definition 2.14. A function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ is *complex differentiable* or has a *complex derivative* at the point $a \in \mathcal{D}$ iff the limit

$$\lim_{z \to a; z \in \mathcal{D} \setminus \{a\}} \frac{f(z) - f(a)}{z - a} \tag{2.6}$$

exists, in which case we denote the limit by $f'(a)$.

Lemma 2.15. Consider a function $f(x) : \mathbb{R} \to \mathbb{C}$. It is complex differentiable if and only if its real and imaginary parts, $u(x)$ and $v(x)$, are differentiable. If so we have

$$f'(x) = u'(x) + \mathbf{i}v'(x). \tag{2.7}$$

Definition 2.16. A point $a \in \mathbb{C}$ is called an *accumulation point* or a *limit point* or a *cluster point* of $\mathcal{D} \subset \mathbb{C}$ iff

$$\forall \epsilon > 0, \exists z \in \mathcal{D} \text{ s.t. } |z - a| \in (0, \epsilon). \tag{2.8}$$

Definition 2.17. An *adherent point*, or a *closure point* or a *point of closure* or a *contact point*, of a subset A of a topological space $\mathcal{X}$, is a point $x \in \mathcal{X}$ such that every open set containing x contains at least one point of A.

Lemma 2.18. Let $a \in \mathcal{D}$ be an accumulation point of $\mathcal{D} \subset \mathbb{C}$, $f : \mathcal{D} \to \mathbb{C}$ a function, and $\ell \in \mathbb{C}$. The following statements are equivalent:

(a) f is complex differentiable at a with $f'(a) = \ell$.

(b) There exists a function $\varphi : \mathcal{D} \to \mathbb{C}$ which is continuous at a such that $\varphi(a) = \ell$ and

$$f(z) = f(a) + \varphi(z)(z - a).$$

(c) There exists a function $\rho : \mathcal{D} \to \mathbb{C}$ which is continuous at a such that $\rho(a) = 0$ and

$$f(z) = f(a) + \ell(z - a) + \rho(z)(z - a).$$

(d) If one defines $r : \mathcal{D} \to \mathbb{C}$ by

$$f(z) = f(a) + \ell(z - a) + r(z),$$

then we have $\lim_{z \to a; z \in \mathcal{D} \setminus \{a\}} \frac{r(z)}{z-a} = 0$.

Proof. (a) $\implies$ (b). (a) and Definition 2.14 imply

$$\lim_{z \to a; z \in \mathcal{D} \setminus \{a\}} \frac{f(z) - f(a)}{z - a} = \ell.$$

By Definition 2.1, the function $\varphi : \mathcal{D} \to \mathbb{C}$ below is continuous at a:

$$\varphi(z) := \begin{cases} \ell & \text{if } z = a; \\ \frac{f(z)-f(a)}{z-a} & \text{otherwise.} \end{cases}$$

(b) $\implies$ (c) is given by

$$\rho(z) := \begin{cases} 0 & \text{if } z = a; \\ \varphi(z) - \ell & \text{otherwise.} \end{cases}$$

(c) $\implies$ (d) is given by the continuity of ρ and

$$r(z) := f(z) - f(a) - \ell(z - a) = (z - a)\rho(z).$$

(d) $\implies$ (a) is given by Definition 2.14. □

Corollary 2.19. A function that is differentiable at a is continuous at a.

Theorem 2.20. Let the functions $f, g : \mathcal{D} \to \mathbb{C}$, $\mathcal{D} \subseteq \mathbb{C}$ be complex differentiable at $a \in \mathcal{D}$. Then the functions λf with $\lambda \in \mathbb{C}$, $\frac{1}{f}$ with $f(a) \neq 0$, $f + g$, and $f \cdot g$ are complex differentiable at a, and we have

$$(\lambda f)'(a) = \lambda f'(a), \tag{2.9}$$

$$\left(\frac{1}{f}\right)'(a) = -\frac{f'}{f^2}(a), \tag{2.10}$$

$$(f + g)'(a) = f'(a) + g'(a), \tag{2.11}$$

$$(fg)'(a) = f'(a)g(a) + f(a)g'(a). \tag{2.12}$$

Exercise 2.21. Prove Theorem 2.20.

Exercise 2.22. Show that the function $f : \mathcal{D} \to \mathbb{C}$ given by

$$f(z) = z^n \text{ with } \begin{cases} \mathcal{D} = \mathbb{C} & \text{if } n \in \mathbb{N}, \\ \mathcal{D} = \mathbb{C}^\bullet & \text{if } n \in -\mathbb{N}^+ \end{cases}$$

is complex differentiable and $f'(z) = nz^{n-1}$. Then deduce that the complex derivative of the complex polynomial (2.2) is

$$p_n'(z) = \sum_{j=1}^{n} j a_j z^{j-1}. \tag{2.13}$$

Example 2.23. For complex polynomials $p, q : \mathbb{C} \to \mathbb{C}$ with the affine variety $\mathcal{N}(q) := \{z \in \mathbb{C} : q(z) = 0\}$, the function

$$z \mapsto \frac{p(z)}{q(z)} : \quad \mathbb{C} \setminus \mathcal{N} \to \mathbb{C} \tag{2.14}$$

is complex differentiable and its derivative follows directly from (2.10), (2.12), and (2.13).

Theorem 2.24. The complex exponential function is complex differentiable on $\mathbb{C}$ with its derivative as itself.

Proof. By Definition 1.83, we have

$$\frac{\exp z - 1}{z} - 1 = \sum_{n=1}^{+\infty} \frac{z^n}{(n+1)!}; \quad \frac{\exp |z| - 1}{|z|} - 1 = \sum_{n=1}^{+\infty} \frac{|z|^n}{(n+1)!},$$

which, together with Corollary 1.66 (1,2), imply

$$\left| \frac{\exp z - 1}{z} - 1 \right| \leq \frac{\exp |z| - 1}{|z|} - 1. \tag{2.15}$$

Take the limit $\lim_{z \to 0}$ on both sides of (2.15), apply the known result on $\exp : \mathbb{R} \to \mathbb{R}$, and we have

$$\lim_{z \to 0; z \in \mathbb{C}} \frac{\exp z - \exp(0)}{z - 0} = 1, \tag{2.16}$$

which, by Definition 2.14, implies that $z \mapsto \exp(z)$ is differentiable at $z = 0$ with its derivative as 1. Consequently, for any fixed $w \in \mathbb{C}^\bullet$, Theorem 1.89 yields

$$e^{-w} \lim_{z \to w} \frac{e^z - e^w}{z - w} = \lim_{z \to w} \frac{e^{z-w} - e^0}{z - w} = \lim_{y \to 0} \frac{e^y - e^0}{y - 0} = 1,$$

where the second step follows from the variable substitution $y = z - w$ and the last from (2.16). Multiplying both sides by e^w completes the proof. □

Theorem 2.25 (Derivative of inverse functions)**.** Suppose two continuous functions $f : \mathcal{D} \to \mathbb{C}$ and $g : \mathcal{D}' \to \mathbb{C}$ satisfy $f(\mathcal{D}) \subseteq \mathcal{D}'$ and $g(f(z)) = z$ for all $z \in \mathcal{D}$. If g is complex differentiable at $b = f(a)$ and $g'(b) \neq 0$, then f is complex differentiable at a and we have

$$f'(a) = \frac{1}{g'(b)}. \tag{2.17}$$

Proof. By Definition 2.14 and the given conditions we have

$$\begin{aligned}\lim_{z\to a} \frac{f(z) - f(a)}{z - a} &= \lim_{z\to a} \frac{f(z) - f(a)}{g(f(z)) - g(f(a))} \\ &= \frac{1}{\lim_{z\to a} \frac{g(f(z))-g(f(a))}{f(z)-f(a)}} = \frac{1}{g'(b)}.\end{aligned}$$ □

Corollary 2.26. In addition to Corollary 2.13, the principal value of the logarithm is complex differentiable on $\mathbb{C}_-$ with its derivative as

$$(\mathrm{Log} z)' = \frac{1}{z}. \tag{2.18}$$

Proof. This follows directly from Theorems 2.24, 2.25, and 1.110. □

Theorem 2.27 (Chain rule)**.** Suppose two functions $f : \mathcal{D} \to \mathbb{C}$ and $g : \mathcal{D}' \to \mathbb{C}$ satisfy $f(\mathcal{D}) \subseteq \mathcal{D}'$. If f is complex differentiable at $a \in \mathcal{D}$ and g is complex differentiable at $f(a) \in \mathcal{D}'$, then their composition $g \circ f : \mathcal{D} \to \mathbb{C}$ is differentiable at $z = a$ and

$$(g \circ f)'(a) = g'(f(a))f'(a). \tag{2.19}$$

Proof. By Lemma 2.18 (b) and the condition of f and g being complex differentiable, there exist functions φ and ψ respectively continuous at a and $b = f(a)$ such that

$$\begin{aligned} f(z) - f(a) &= \varphi(z)(z - a), \quad \varphi(a) = f'(a); \\ g(w) - g(b) &= \psi(w)(w - b), \quad \psi(b) = g'(b). \end{aligned}$$

It follows that $\frac{g(f(z))-g(f(a))}{z-a} = \psi(f(z))\frac{f(z)-f(a)}{z-a}$. Take the limit $z \to a$ and we have

$$(g \circ f)'(a) = \psi(f(a))f'(a) = g'(f(a))f'(a).$$ □

Corollary 2.28. The sine and cosine functions are complex differentiable and

$$\forall z \in \mathbb{C}, \qquad \sin' z = \cos z, \qquad \cos' z = -\sin z. \tag{2.20}$$

Proof. This follows directly from Theorem 2.24, Theorem 2.27, and Corollary 1.94. □

Corollary 2.29. For any given $s \in \mathbb{C}$, the power function $f : \mathbb{C}_- \to \mathbb{C}$ defined as

$$f(z) = z^s := \exp(s\mathrm{Log} z) \tag{2.21}$$

is complex differentiable with its derivative as

$$f'(z) = \exp(s\mathrm{Log} z) \cdot s\frac{1}{z} = sz^{s-1}. \tag{2.22}$$

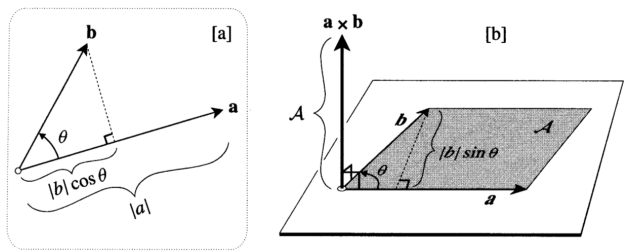
Proof. This follows directly from Theorem 2.27, Theorem 2.24, and Corollary 2.26. □

2.3 Analytic = conformal

Definition 2.30. A function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ being open is called *analytic* or *holomorphic* or *regular* in $\mathcal{D}$ iff it is complex differentiable at every point of $\mathcal{D}$. In particular, f is called *analytic at a point* $a \in \mathcal{D}$ iff there exists an open neighborhood $\mathcal{U} \subset \mathcal{D}$ of a such that f is analytic in $\mathcal{U}$.

Definition 2.31. For $z = x + \mathbf{i}y$, $w = u + \mathbf{i}v$ with $x, y, u, v \in \mathbb{R}$, the *standard scalar product* in $\mathbb{C}$ with respect to the basis $(1, \mathbf{i})$ is defined by

$$\langle z, w\rangle := \mathrm{Re}\,(\overline{z}w) = xu + yv. \tag{2.23}$$



Exercise 2.32. Define a Hermitian inner product $(\cdot, \cdot)$ in $\mathbb{C}$ by

$$(z, w) := \overline{z}w.$$

The term Hermitian is used to discribe the fact that $(\cdot, \cdot)$ is not symmetric, but rather satisfies the relation

$$\forall z, w \in \mathbb{C}, \quad (z, w) = \overline{(w, z)}.$$

Show that

$$\langle z, w\rangle = \frac{1}{2}[(z, w) + (w, z)] = \mathrm{Re}\,(z, w). \tag{2.24}$$

Lemma 2.33. For each pair $(z, w) \in \mathbb{C}^\bullet \times \mathbb{C}^\bullet$, there exists a unique real number $\theta := \theta(z, w) \in (-\pi, \pi]$, called the *oriented angle* from z to w, such that

$$\begin{cases} \cos\theta = \cos\theta(z, w) = \dfrac{\langle z, w\rangle}{|z||w|}, \\ \sin\theta = \sin\theta(z, w) = \dfrac{\langle \mathbf{i}z, w\rangle}{|z||w|}. \end{cases} \tag{2.25}$$

Proof. Definition 2.31 yields

$$\forall z, w \in \mathbb{C}, \qquad \langle z, w\rangle^2 + \langle \mathbf{i}z, w\rangle^2 = |z|^2|w|^2, \tag{2.26}$$

hence both $\frac{\langle z,w\rangle}{|z||w|}$ and $\frac{\langle \mathbf{i}z,w\rangle}{|z||w|}$ are in the range of $[-1, 1]$. It can also be verified by direct calculation that

$$\frac{\overline{z}w}{|z||w|} = \frac{\langle z, w\rangle}{|z||w|} + \mathbf{i}\frac{\langle \mathbf{i}z, w\rangle}{|z||w|}. \tag{2.27}$$

By Definitions 1.61 and 1.108, $\mathrm{Arg}\frac{w}{z}$ is the oriented angle from z to w and is uniquely determined by (2.25). □

Corollary 2.34. The *law of cosines* in trigonometry can be expressed as

$$|z + w|^2 = |z|^2 + 2\,\langle z, w\rangle + |w|^2, \tag{2.28a}$$

$$|z - w|^2 = |z|^2 - 2\,\langle z, w\rangle + |w|^2, \tag{2.28b}$$

where z, w are arbitrary nonzero complex numbers.

Definition 2.35. An $\mathbb{R}$-linear operator $T \in \mathcal{L}(\mathbb{R}^2)$ is *orientation preserving* iff $\det T > 0$.

Definition 2.36. An $\mathbb{R}$-linear operator $T \in \mathcal{L}(\mathbb{R}^2)$ is *angle-preserving* iff it satisfies

$$\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \quad |T\mathbf{x}||T\mathbf{y}| \langle \mathbf{x}, \mathbf{y} \rangle = |\mathbf{x}||\mathbf{y}| \langle T\mathbf{x}, T\mathbf{y} \rangle, \tag{2.29}$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product in Definition 2.31 and $|\cdot|$ the Euclidean norm.

Example 2.37. The linear map $z \mapsto \overline{z}$ is angle-preserving but not orientation preserving.

Lemma 2.38. A linear operator $T \in \mathcal{L}(\mathbb{R}^2)$ preserves oriented angles if and only if it is a normal operator as defined in (1.53).

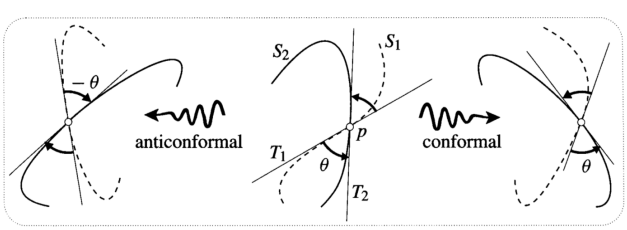
Proof. Suppose the matrix of T is an element of $\mathbb{R}_N^{2\times 2}$ as defined in (1.53). It is then straightforward to verify Definitions 2.35 and 2.36.

Now suppose T is both angle and orientation preserving. Write the matrix of T as

$$M(T) = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Set $\mathbf{x} = (1, 0)^\top$ and $\mathbf{y} = (0, 1)^\top$, plug them into (2.29), and we have $ab + cd = 0$. Set $\mathbf{x} = (1, 1)^\top$ and $\mathbf{y} = (1, -1)^\top$, plug them into (2.29), and we have $a^2 + c^2 = b^2 + d^2$, the substitution of $a = -\frac{cd}{b}$ into which yields $a^2 = d^2$ and $b^2 = c^2$. Suppose $b = c$, then $a = -d$ and $\det T < 0$, a contradiction to T being orientation preserving. Hence we have $b = -c$, $a = d$, and $a^2 + b^2 \neq 0$, which complete the proof. □

Definition 2.39 (Conformal and anticonformal maps)**.** Let $\mathcal{D}, \mathcal{D}' \subseteq \mathbb{R}^n$ be open. Let $f : \mathcal{D} \to \mathcal{D}'$ be a continuously differentiable map. f is said to be (locally) *conformal* iff its Jacobian matrix $J(f; a)$ is both angle and orientation preserving at each point $a \in \mathcal{D}$. f is *anticonformal* iff it preserves the magnitude of the angle but reverses the orientation at each point $a \in \mathcal{D}$. f is *globally conformal* if it is bijective and conformal.



Theorem 2.40. A $\mathcal{C}^1$ function $f : \mathcal{D} \to \mathcal{D}'$ with $\mathcal{D}, \mathcal{D}' \subseteq \mathbb{R}^2$ open is a conformal mapping if and only if f is analytic and $f' \neq 0$ when $\mathcal{D}, \mathcal{D}'$ are interpreted as open subsets of $\mathbb{C}$.

Proof. Suppose f is conformal. By the continuous differentiability of f, Definition 2.39 and Lemmas 2.38 and 1.122, the action of the Jacobian $J(f; a)$ is to multiply the infinitesimal vector $\lim_{z\to a}(z - a)$ with a constant $c \in \mathbb{C}$:

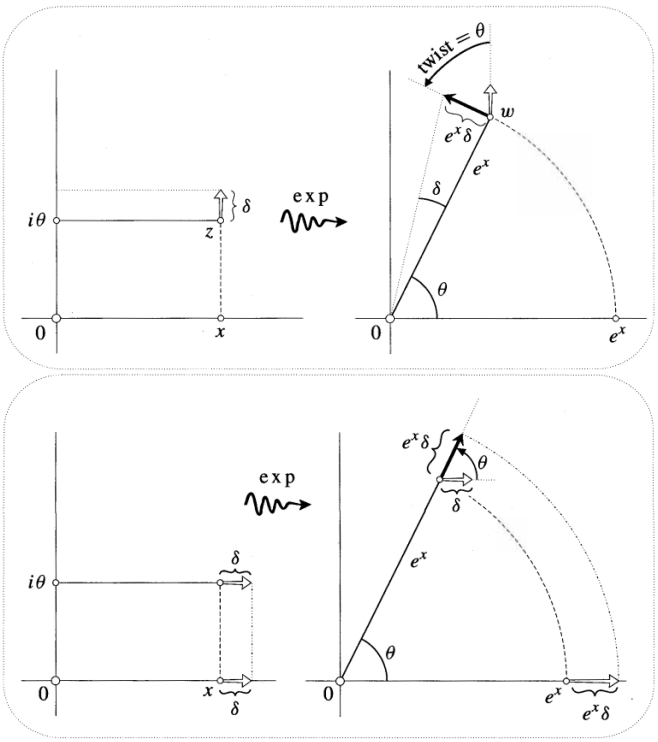
$$\lim_{z\to a}(f(z) - f(a)) = \lim_{z\to a} J(f; a)(z - a) = c \lim_{z\to a}(z - a),$$

which implies that f is analytic with $f'(a) = c$.

The sufficiency follows directly from Lemma 1.122 and the fact that a $\mathbb{C}$-linear map can always be interpreted as an $\mathbb{R}$-linear map. □

Example 2.41. Consider the exponential function $z \mapsto e^z$ in Definition 1.83. By Theorem 2.24 and Definition 2.30, $z \mapsto e^z$ is analytic. By Theorem 2.40 and Definition 2.39, $z \mapsto e^z$ is locally conformal from $\mathbb{C}$ to $\mathbb{C}$ and globally conformal from $\{z \in \mathbb{C} : \mathrm{Im}(z) \in (-\pi, \pi)\}$ to $\mathbb{C}_-$. The behavior of $z \mapsto e^z$ in Example 1.86 is well understood by Definitions 2.35, 2.36, and 2.39.

Example 2.42 (Visual differentiation of the exponential function)**.** The derivative of the exponential function $z \mapsto e^z$ can be obtained via a geometric reasoning: it equals the complex number ℓ that satisfies $\exp(z + \delta) - \exp(z) = \ell\delta$.



Exercise 2.43. Give a visual differentiation of the principle branch of the logrithm function defined in Theorem 1.110. You must draw two plots similar to those in Example 2.42 and elaborate on how the derivative is derived from the plots.

Exercise 2.44. Repeat Exercise 2.43 for the power function with $n \in \mathbb{Z}^+$, as defined in (1.50).

2.4 Cauchy–Riemann equations

Definition 2.45. A function $f : \mathcal{D} \to \mathbb{R}^q$ with $\mathcal{D} \subseteq \mathbb{R}^p$ being open is *totally differentiable* at a point $a \in \mathcal{D}$ iff there exists an $\mathbb{R}$-linear map $A : \mathbb{R}^p \to \mathbb{R}^q$ such that the remainder r introduced by the equation

$$f(x) - f(a) = A(x - a) + r(x)$$

satisfies

$$\lim_{x\to a} \frac{r(x)}{|x - a|} = 0,$$

where $|x - a|$ is the Euclidean distance between x and a. A is called the *Jacobian* or *total differential* of f at a.

Definition 2.46. The *partial derivatives* of a complex function $f(z) = f(x + \mathbf{i}y) = u(x, y) + \mathbf{i}v(x, y)$ are

$$\frac{\partial f}{\partial x} = \frac{\partial u}{\partial x} + \mathbf{i}\frac{\partial v}{\partial x}, \qquad \frac{\partial f}{\partial y} = \frac{\partial u}{\partial y} + \mathbf{i}\frac{\partial v}{\partial y}, \tag{2.30}$$

Lemma 2.47. A function $\mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ being open is complex differentiable at $a \in \mathcal{D}$ if and only if f is totally differentiable at a (in the sense of Definition 2.45) and its Jacobian $J(f) : \mathcal{D} \to \mathbb{C}$ is of the form

$$J(f; a)(z - a) = \ell(z - a), \tag{2.31}$$

where ℓ, the derivative of f, is a constant complex number.

Proof. For necessity, f being complex differentiable, Definition 2.14, Lemma 2.18(a,d), and the isomorphism $\mathbb{C} \simeq \mathbb{R}^2$ in Theorem 1.50 imply that f is totally differentiable in the sense of Definition 2.45. In particular, the $\mathbb{C}$-linear map ℓ in Lemma 2.18(a,d) can be regarded as the $\mathbb{R}$-linear map A in Definition 2.45. In addition, (2.31) follows trivially from f being complex differentiable as $\ell = f'(a)$.

For sufficiency, $J(f; a)$ being an $\mathbb{R}$-linear map, (2.31), and Lemma 1.122 implies that $J(f; a)$ is also a $\mathbb{C}$-linear map. Then the proof is completed by the isomorphism in Theorem 1.121. □

Corollary 2.48. Let $\mathcal{D} \subseteq \mathbb{C}$ be open and $f : \mathcal{D} \to \mathbb{C}$ be a complex differentiable function with continuous derivative. If $f'(a) \neq 0$ at a point $a \in \mathcal{D}$, then there exists an open set $\mathcal{D}_0 \subseteq \mathcal{D}$ with $a \in \mathcal{D}_0$ such that the restriction $f|_{\mathcal{D}_0}$ is injective and $\forall z \in \mathcal{D}_0$, $f'(z) \neq 0$.

Proof. By $f'(a) \neq 0$ and the continuity of $f'(z)$, there exists an open neighborhood $\mathcal{D}_0$ of a such that $\forall z \in \mathcal{D}_0$, $f'(z) \neq 0$. By Lemma 2.47, $f'(a)(z - a) = J(f; a)(z - a)$. Then the condition $f'(a) \neq 0$ implies that $J(f; a)$ in the form of (1.57) is locally a bijection. □

Theorem 2.49 (Cauchy–Riemann)**.** Let $\mathcal{D} \subseteq \mathbb{C}$ be open and $a \in \mathcal{D}$. The following statements are equivalent for a function $f : \mathcal{D} \to \mathbb{C}$:

(a) f is complex differentiable at a.

(b) f is totally differentiable at a in the sense of real analysis ($\mathbb{C} = \mathbb{R}^2$), and for $u := \operatorname{Re} f$ and $v := \operatorname{Im} f$ the following differential equations hold

$$\frac{\partial u}{\partial x}(a) = \frac{\partial v}{\partial y}(a), \qquad \frac{\partial u}{\partial y}(a) = -\frac{\partial v}{\partial x}(a). \tag{2.32}$$

In case that (a) or (b) holds one has

$$f'(a) = \frac{\partial u}{\partial x}(a) + \mathbf{i}\frac{\partial v}{\partial x}(a) = \frac{\partial v}{\partial y}(a) - \mathbf{i}\frac{\partial u}{\partial y}(a). \tag{2.33}$$

Proof. This follows from Lemmas 2.47 and 1.122. □

Notation 2. Instead of $\frac{\partial u}{\partial x}(a)$ and $\frac{\partial u}{\partial y}(a)$, one often writes $\partial_x u(a)$ and $\partial_y u(a)$, or, $\partial_1 u(a)$ and $\partial_2 u(a)$.

Corollary 2.50. A complex function $f : \mathcal{D} \to \mathbb{C}$ differentiable at $a \in \mathcal{D}$ satisfies

$$\det J(f; a) = |f'(a)|^2 = (\partial_x u)^2 + (\partial_x v)^2 = (\partial_y u)^2 + (\partial_y v)^2. \tag{2.34}$$

Exercise 2.51. Prove Corollary 2.50.

Example 2.52. The complex conjugate function $f(z) = \overline{z}$ is not complex differentiable anywhere since $1 = \partial_1 u \neq \partial_2 v = -1$.

Example 2.53. Theorem 2.24 is easily verified by the Cauchy–Riemann conditions.

Example 2.54. The function $f : \mathbb{C} \to \mathbb{C}$ given by

$$f(z) = \begin{cases} 0 & \text{if } z = 0, \\ \frac{x^3 y(y - \mathbf{i}x)}{x^6 + y^2} & \text{otherwise.} \end{cases} \tag{2.35}$$

satisfies the Cauchy–Riemann equations, but is not totally differentiable: taking $z = t + \mathbf{i}t^3$ and $z = te^{\mathbf{i}\theta}$ for $t \in \mathbb{R}$ and fixed $\theta \in \mathbb{R}$ yields different limits. Hence $f(z)$ is not analytic at $z = 0$.

Corollary 2.55. A function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ being open is locally constant in $\mathcal{D}$ if and only if f is complex differentiable on $\mathcal{D}$ and $f'(z) = 0$ for any $z \in \mathcal{D}$.

Proof. The necessity follows directly from Definition 2.14. For sufficiency, write $f(x + \mathbf{i}y) = u(x, y) + \mathbf{i}v(x, y)$. Then $f'(z) = 0$ and Theorem 2.49 imply that $\partial_x u(a) = \partial_x v(a) = 0$ and $\partial_y u(a) = \partial_y v(a) = 0$. Hence both $u(x, y)$ and $v(x, y)$ are locally constant, so is f. □

Corollary 2.56. Any complex differentiable function in $\mathcal{D}$ with only real (or purely imaginary) values is locally constant in $\mathcal{D}$.

Exercise 2.57. Prove Corollary 2.56.

Example 2.58. It follows from Corollary 2.56 that the functions $f(z) = |\sin z|$ and $f(z) = \operatorname{Re} z$ are not complex differentiable in $\mathbb{C}$.

Corollary 2.59. A subset $\mathcal{D} \subseteq \mathbb{C}$ is connected if and only if each locally constant function $f : \mathcal{D} \to \mathbb{C}$ is constant.

Exercise 2.60. Prove Corollary 2.59.

Corollary 2.61. The real part of an analytic function in a connected open set $\mathcal{D} \subseteq \mathbb{C}$ is uniquely determined by its imaginary part, up to an additive constant.

Exercise 2.62. Prove Corollary 2.61.

Exercise 2.63. Suppose an analytic function $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ being open satisies $|f|$ is constant in $\mathcal{D}$. Prove that f is constant in $\mathcal{D}$.

Definition 2.64. For a vector space of analytic functions, the *linear operators of partial differentiation with respect to* $z = x + \mathbf{i}y$ *and* $\overline{z} = x - \mathbf{i}y$ are

$$\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - \mathbf{i}\frac{\partial}{\partial y}\right), \quad \frac{\partial}{\partial \overline{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + \mathbf{i}\frac{\partial}{\partial y}\right). \tag{2.36}$$

Theorem 2.65. A complex function $f(z) = u(x,y) + \mathbf{i}v(x,y)$ is analytic on an open set $\mathcal{D} \subseteq \mathbb{C}$ if and only if $(u,v)^\top : \mathbb{R}^2 \to \mathbb{R}^2$ is totally differentiable and

$$\forall z \in \mathcal{D}, \qquad \frac{\partial f}{\partial \bar{z}}(z) = 0, \tag{2.37}$$

in which case $f' = \frac{\partial f}{\partial z}$.

Proof. (2.37) follows from Definition 2.64 and Theorem 2.49 while $f' = \frac{\partial f}{\partial z}$ from (2.33) and (2.36). □

Exercise 2.66. Show that in polar coordinates, the Cauchy–Riemann equations take the form

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}, \qquad \frac{1}{r}\frac{\partial u}{\partial \theta} = -\frac{\partial v}{\partial r}.$$

Use these equations to show that the logarithm function defined by

$$\mathrm{Log} z = \log r + \mathbf{i}\theta, \quad z = re^{\mathbf{i}\theta},$$

is analytic in the region $r > 0$ and $-\pi < \theta < \pi$.

2.5 Harmonic functions

Definition 2.67. A $\mathcal{C}^2$ function $u : \mathcal{D} \to \mathbb{R}$ with $\mathcal{D} \subseteq \mathbb{R}^2$ being open is called a *potential function* or a *harmonic function* if it satisfies the *Laplace equation*

$$\Delta u := \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) u = 0, \tag{2.38}$$

where Δ is called the *Laplace operator*.

Exercise 2.68. Show that

$$4\frac{\partial}{\partial z}\frac{\partial}{\partial \bar{z}} = 4\frac{\partial}{\partial \bar{z}}\frac{\partial}{\partial z} = \Delta.$$

Definition 2.69. A harmonic function $v : \mathcal{D} \to \mathbb{R}$ with $\mathcal{D} \subseteq \mathbb{R}^2$ being open is called the *conjugate harmonic function* to a harmonic function $u : \mathcal{D} \to \mathbb{R}$ iff the complex function $f = u + \mathbf{i}v$ is analytic in $\mathcal{D}$.

Theorem 2.70. If $f : \mathcal{D} \to \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ being open is an analytic function whose real and imaginary parts are continuously partial differentiable at least twice, then both the real part and the imaginary part are harmonic functions.

Proof. By the Cauchy–Riemann Theorem 2.49, we have $\partial_1 u = \partial_2 v$ and $\partial_2 u = -\partial_1 v$, which imply $\partial_1^2 u = \partial_1(\partial_2 v)$ and $\partial_2^2 u = -\partial_2(\partial_1 v)$. Partial derivatives commute and hence u is harmonic. Similarly, v is also harmonic. □

Corollary 2.71. If a function $u : \mathcal{D} \to \mathbb{R}$ is harmonic, then the following function $g : \mathcal{D} \to \mathbb{R}$ is analytic:

$$g = \partial_1 u - \mathbf{i}\partial_2 u. \tag{2.39}$$

Proof. This follows directly from Definition 2.69 and Theorem 2.49. □

Theorem 2.72. Suppose $u : \mathcal{D} \to \mathbb{R}$ is a harmonic function where $\mathcal{D} := (a,b) \times (c,d) \subseteq \mathbb{C}$ is an open rectangle whose sides are parallel to the coordinate axes. Then there exists an analytic function $f : \mathcal{D} \to \mathcal{C}$ with u as its real part and with the conjugate harmonic function of u uniquely determined up to an additive constant as

$$v(x,y) := \int_{y_0}^{y} \partial_x u(x,t)\mathrm{d}t - \int_{x_0}^{x} \partial_y u(t,y_0)\mathrm{d}t + v(x_0,y_0), \tag{2.40}$$

where $x_0 \in (a,b)$ and $y_0 \in (c,d)$.

Proof. Define $f = u + \mathbf{i}v$. We claim that f is analytic and that $g = u + \mathbf{i}(v + c)$ is also analytic for any constant $c \in \mathbb{C}$.

By (2.40), we have $\partial_x u = \partial_y v$ and $\partial_y u = -\partial_x v$. Since $u \in \mathcal{C}^2$, $f : \mathbb{R}^2 \to \mathbb{R}^2$ is totally differentiable. Then Theorem 2.49 implies that f is analytic.

By construction, we have $g - f = \mathbf{i}c$. Then the proof is completed by Corollary 2.55. □

Example 2.73. The function

$$u(x,y) := \log\sqrt{x^2 + y^2} \tag{2.41}$$

is harmonic on $\mathbb{C}^\bullet$, but there is no analytic function $f : \mathbb{C}^\bullet \to \mathbb{C}$ that satisfies

$$\mathrm{Re}\, f(z) = \log|z|.$$

We can prove this by contradiction. Suppose there is such a function f. Then by Theorem 1.111 f would have to agree with $g(z) = \mathrm{Log} z + \mathbf{i}c$, where $c \in \mathbb{R}$ is a constant, in $\mathbb{C} \setminus \mathbb{C}_-$, but this contradicts Corollary 2.13, which states that $g(z)$ is not even continuous in $\mathbb{C}^\bullet$.

Exercise 2.74. For an analytic function $f : \mathbb{C}_- \to \mathbb{C}$, prove that $\mathrm{Re}\, f = u$ in (2.41) implies $f(z) = \mathrm{Log}(z) + \mathbf{i}c$ for some constant $c \in \mathbb{R}$.

Exercise 2.75. Find $a \in \mathbb{R}$ such that the function $u_a : \mathbb{R}^2 \to \mathbb{R}$ defined by $u_a(x,y) = x^3 + axy^2$ is harmonic. Furthermore, determine all conjugate harmonic functions to u_a or all analytic functions $f : \mathbb{C} \to \mathbb{C}$ with $\mathrm{Re}\, f = u_a$.